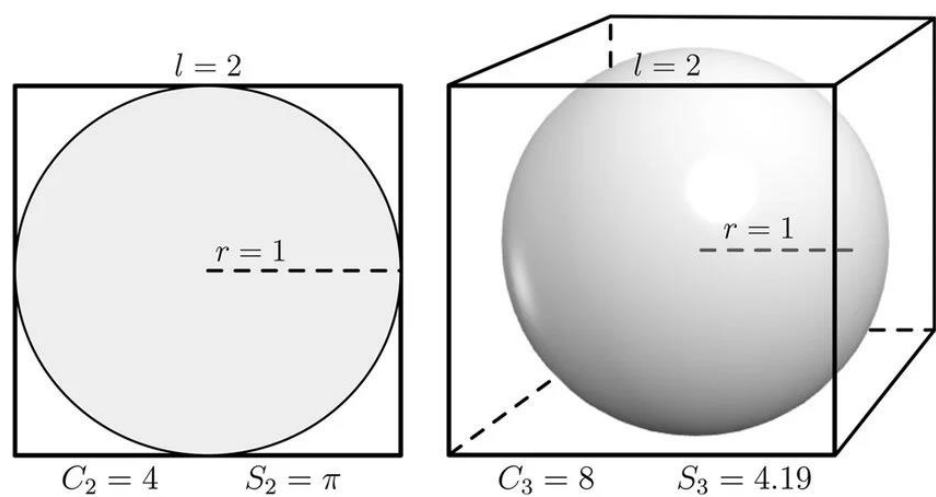


Assignment 4 (part 01)

Perform a Monte Carlo simulation to estimate the value of the constant pi.
It can be done by considering method (a) in two dimensions, inscribe a circle inside of the square or method (b) in three dimensions, inscribe a sphere inside of the cube.

Find out whether method (a) is better than method (b) by considering which one needs more number of runs or replications to achieve a 95% confidence interval with a specified precision $w = 0.01$.



Assignment 4 (part 02)

For either system 1 or system 2, by Monte-Carlo simulation, estimate the probability p that the system will function for at least seven days. Also, find the approximate number of replications required to achieve a 95% confidence interval with a specified precision $w = 0.01$.

System 1

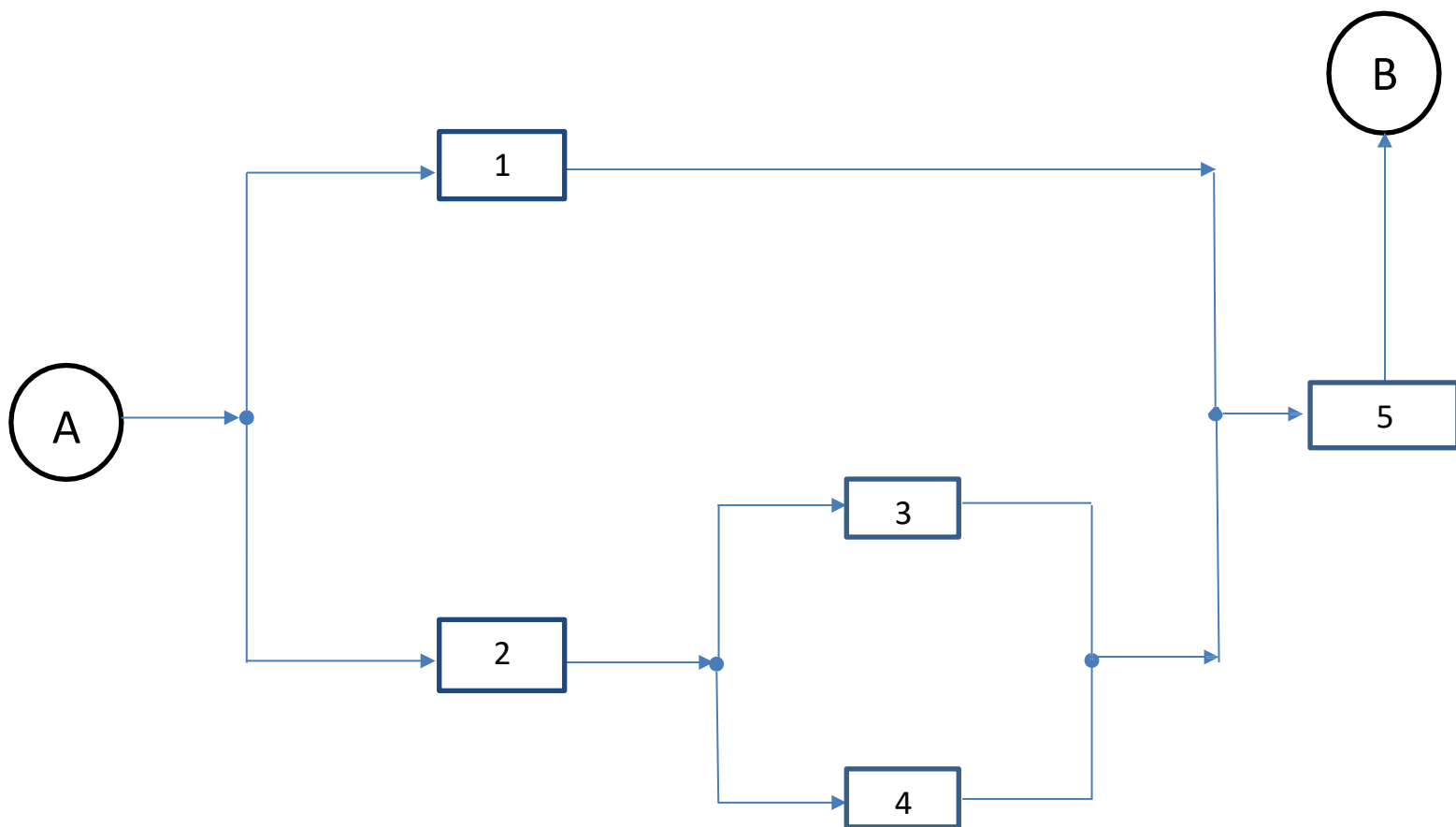
In system 1, it is desired to send a signal from A to B . Each of the five components is subject to random failures, and the components fail independently of each other. Let Y be the time to failure of the entire system and let X_i be the time to failure of component, i , for $i = 1, 2, \dots, 5$. We assume that X_i is exponentially distributed with mean β_i days, with the mean β_i 's given in table below,

i	β_i (in days)
1	10
2	8
3	7
4	5
5	6

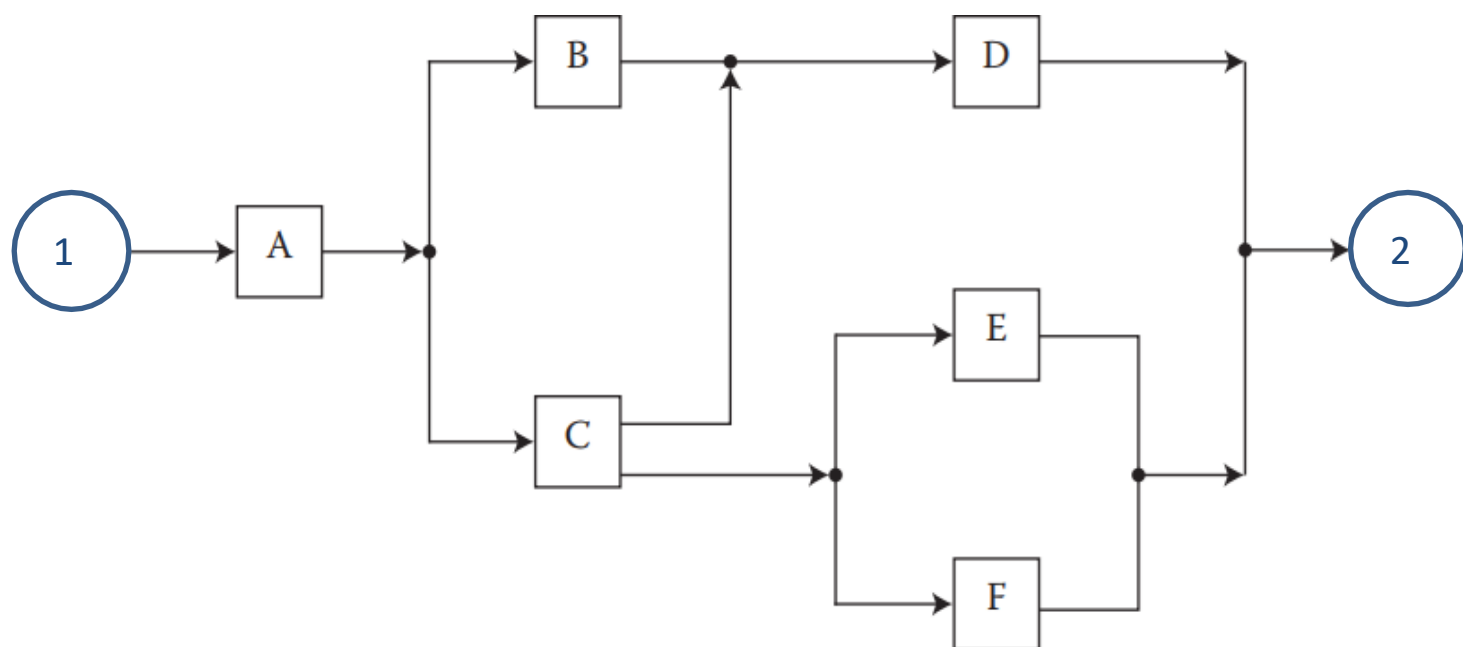
System 2

In system 2, it is desired to send a signal from 1 to 2. Each of the six components is subject to random failures, and the components fail independently of each other. Let Y be the time to failure of the entire system and let X_i be the time to failure of component, i , for $i = 1, 2, \dots, 6$. We assume that X_i is exponentially distributed with mean β_i days, with the mean β_i 's given in table below,

i	β_i (in days)
A	10
B	8
C	7
D	6
E	5
F	4



Five-component stochastic system-1



Six-component stochastic system-2